

Don't Just Listen, Try Planning

Graph-based Retrieval-Generation Agent for Long-form Audio Meeting Understanding

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LongAudioQA Dataset Multi-dimensional Graph Agent Planning Acoustic-aware QA

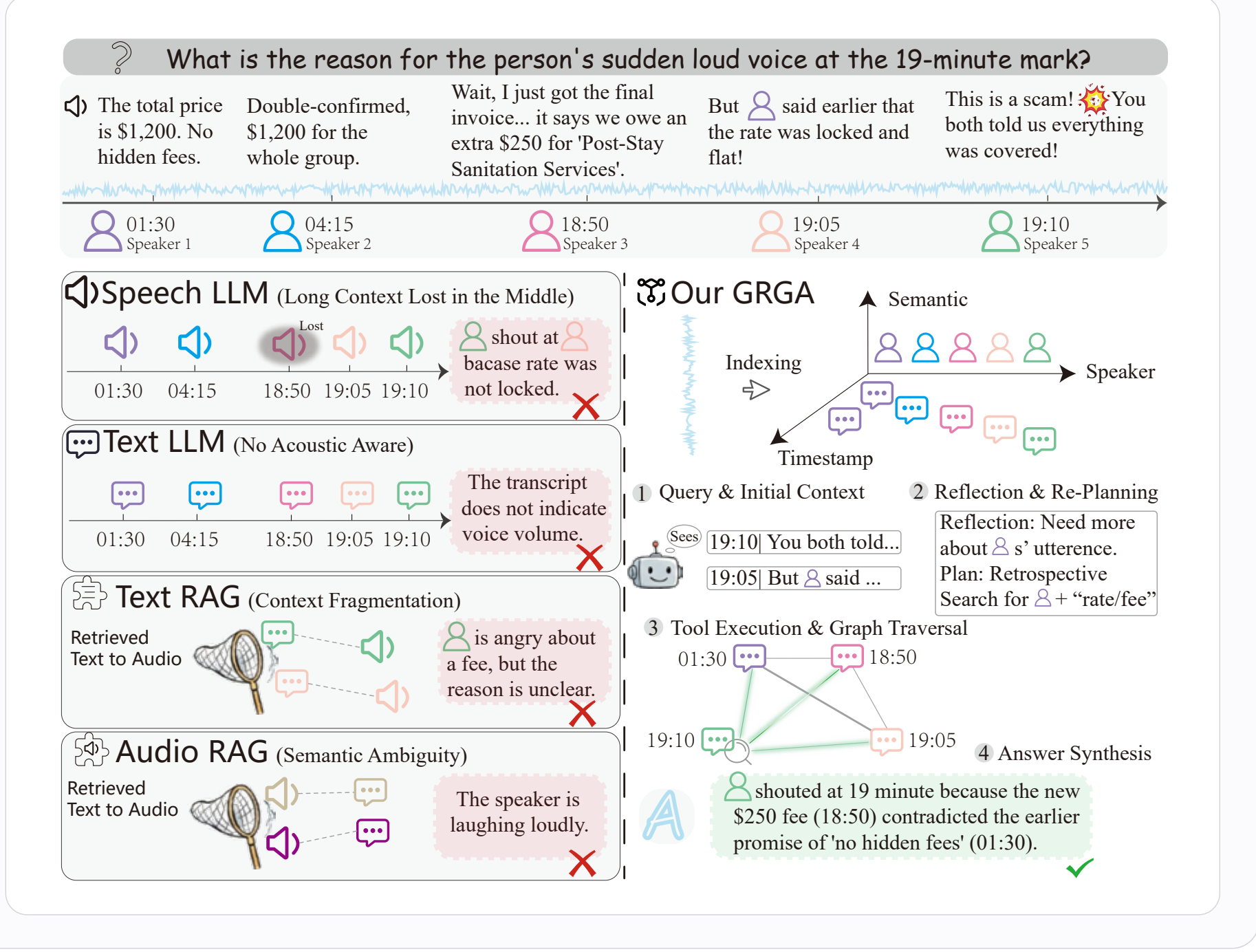
Code & Data

Scan for GRGA resources and LongAudioQA.

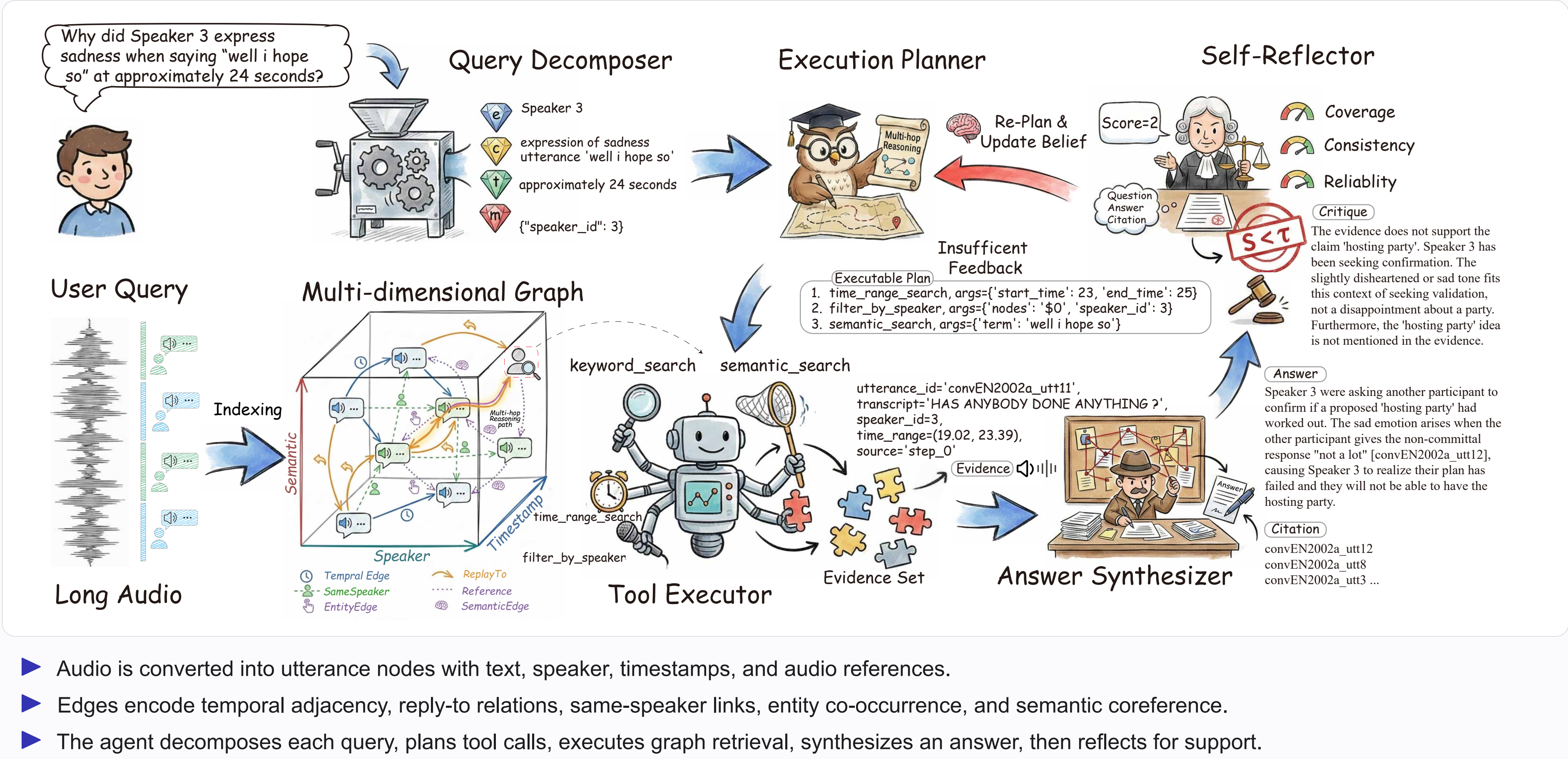
Takeaway:
Plan over semantic, speaker, timestamp, and acoustic evidence instead of retrieving one flat chunk.

Why Long Audio QA Breaks

- ▶ Long-form audio meeting understanding lacks dedicated QA benchmarks.
- ▶ Speech LLMs map audio toward text, losing paralinguistic cues such as loudness, emotion, or speaker tension.
- ▶ Flat RAG retrieves local transcript chunks, but meeting answers often require long-range evidence and speaker/time links.



GRGA: Graph Retrieval + Planning

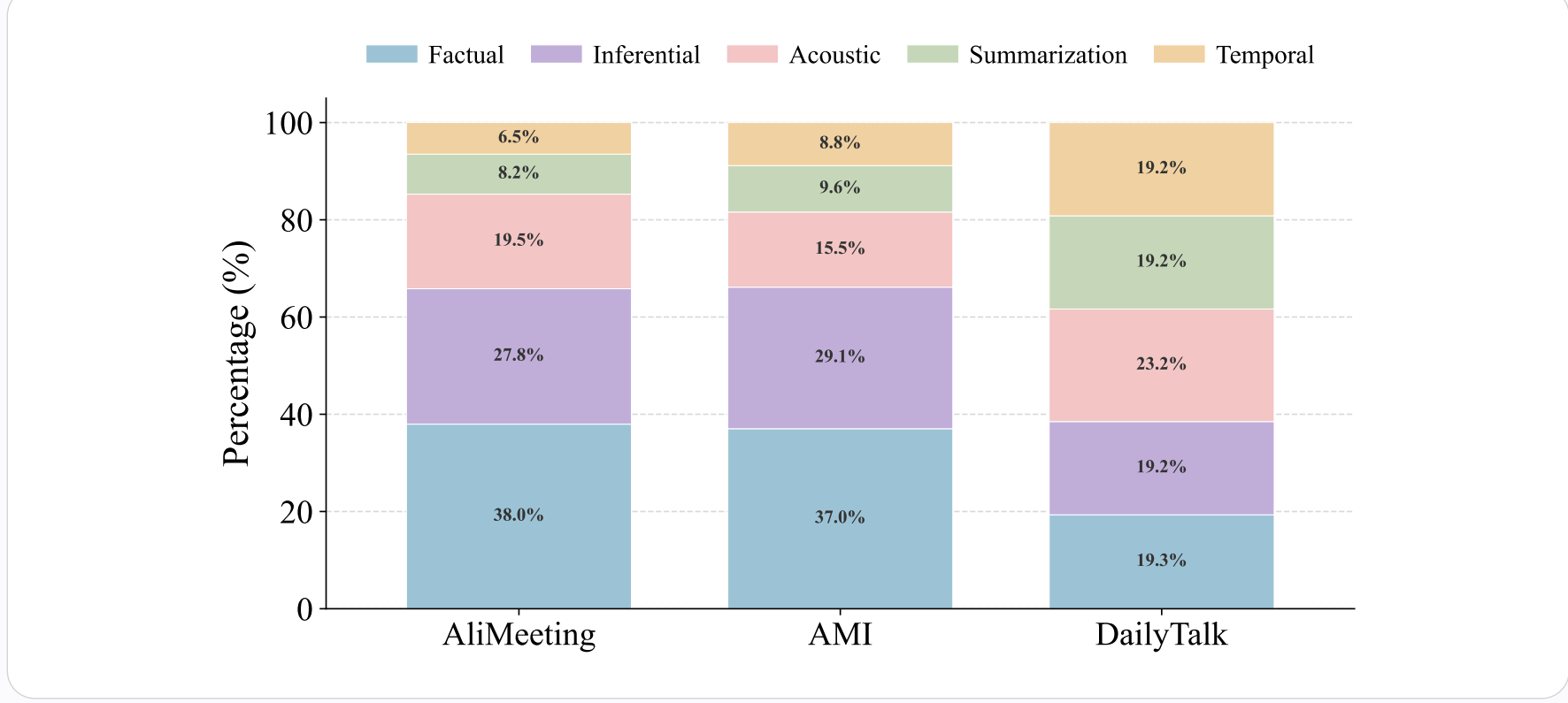


- ▶ Audio is converted into utterance nodes with text, speaker, timestamps, and audio references.
- ▶ Edges encode temporal adjacency, reply-to relations, same-speaker links, entity co-occurrence, and semantic coreference.
- ▶ The agent decomposes each query, plans tool calls, executes graph retrieval, synthesizes an answer, then reflects for support.

LongAudioQA

- ▶ New QA dataset for long-form meeting and conversation understanding.
- ▶ Built from AliMeeting, AMI Meeting, and DailyTalk with human verification.
- ▶ Five skills: factual retrieval, inferential reasoning, temporal grounding, summarization, and acoustic-aware reasoning.

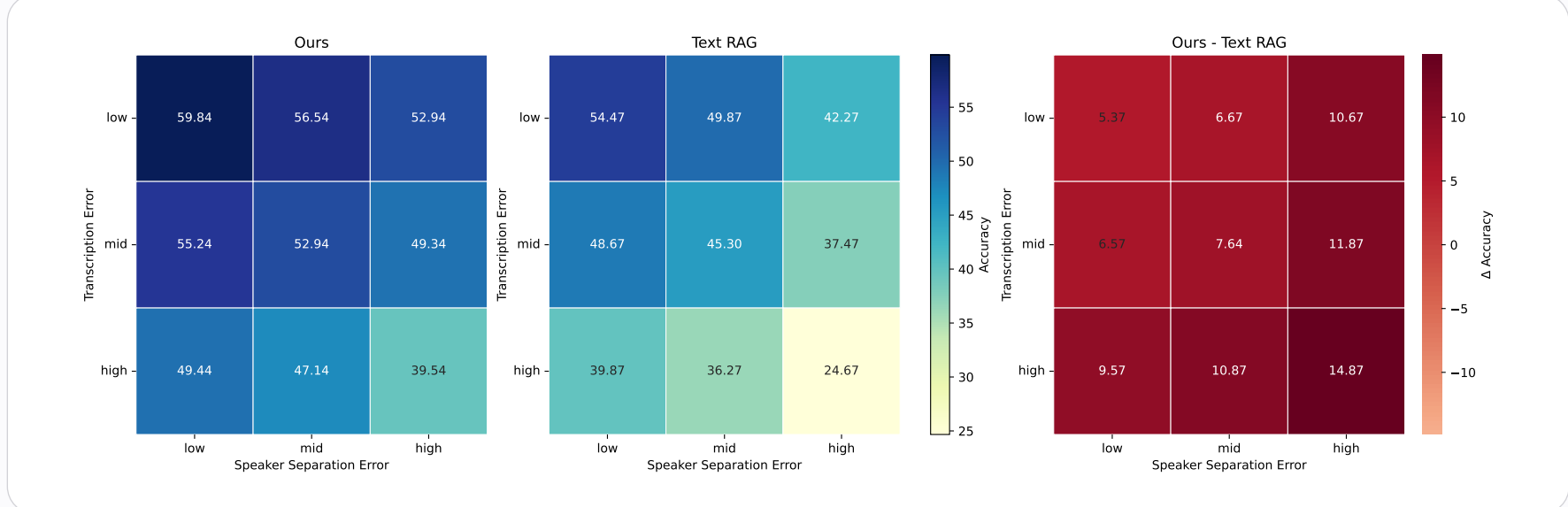
Dataset	Dur(h)	Avg(s)	Turns	Q&A
AliMeeting	14.91	1,935	815	3,013
AMI	18.24	1,930	757	3,243
DailyTalk	21.59	610	186	10,200



Question Taxonomy

Factual	Entity retrieval, attribute association, keyword locating.
Inferential	Multi-hop links between early problems, later discussions, and final decisions.
Temporal	Absolute localization, relative sequencing, frequency statistics in time windows.
Summarization	Topic consensus and speaker contribution aggregation.
Acoustic-aware	Align emotion, intensity, laughter, or volume spikes with semantic context.

Robustness Under Noise



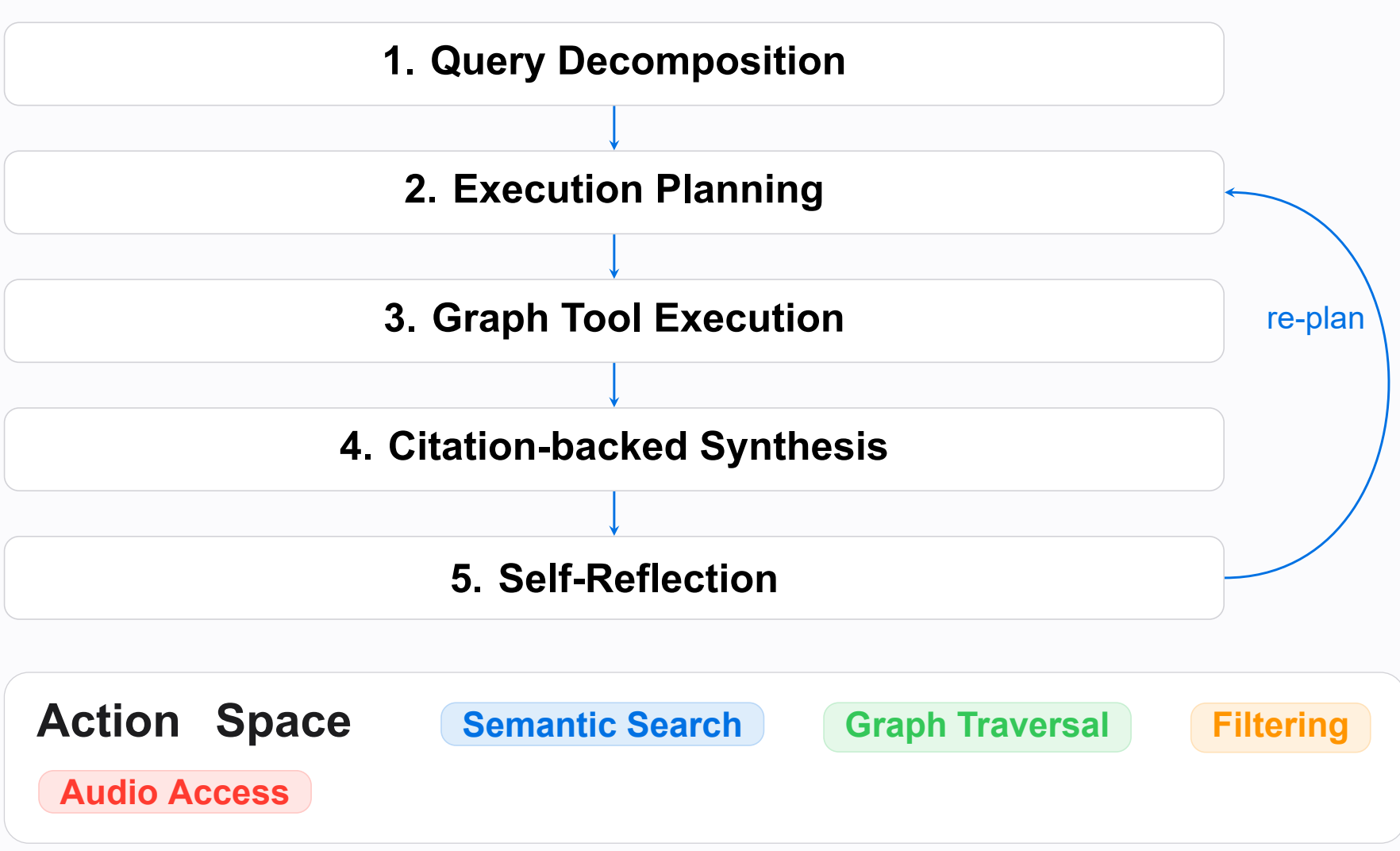
- ▶ Text RAG degrades sharply as ASR and diarization noise increase.
- ▶ GRGA acts as a structured noise buffer by validating evidence through graph links and reflection.

Conclusion

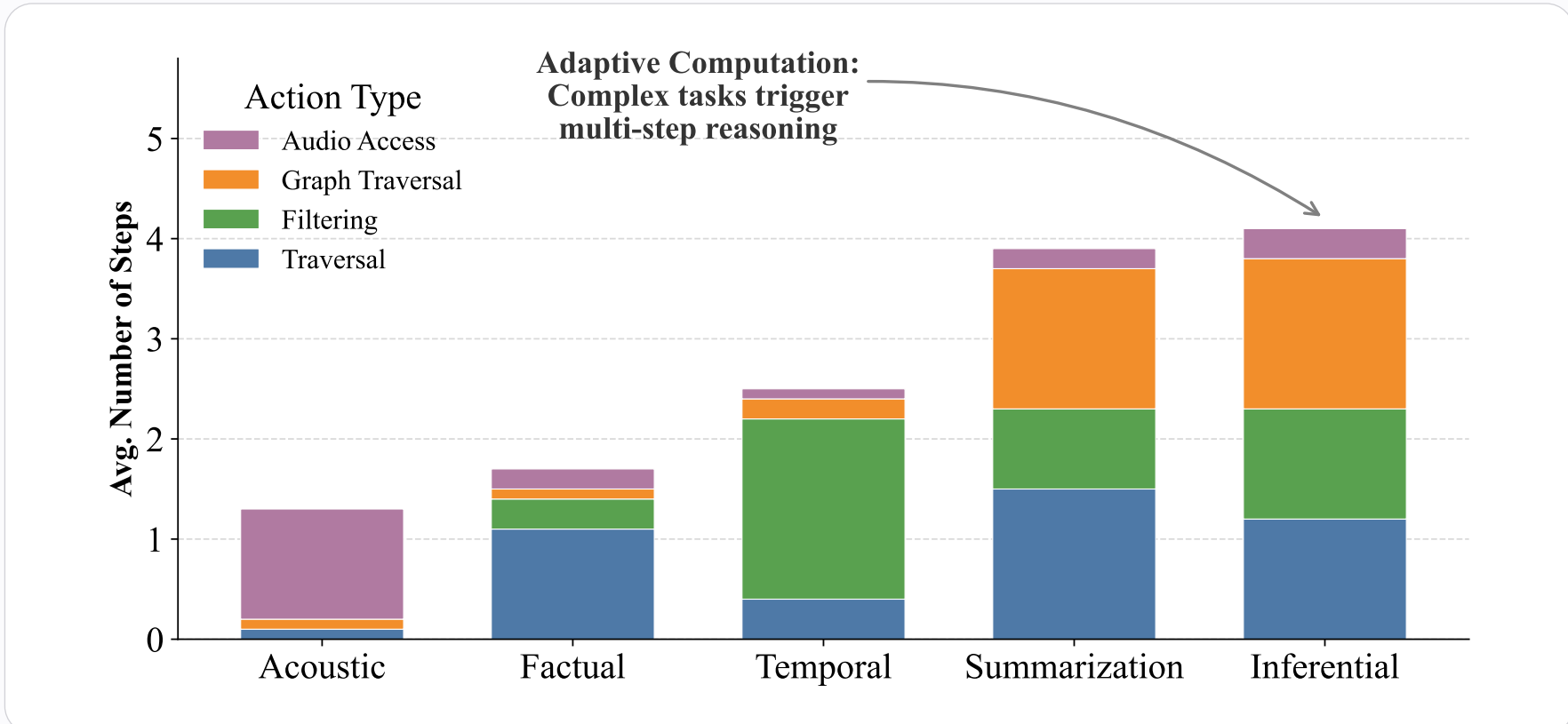
Long-form audio QA needs planning, not just listening.

- ▶ **Dataset:** LongAudioQA introduces long-context, multi-speaker, temporal, and acoustic-aware QA.
- ▶ **Method:** GRGA turns meetings into multi-dimensional graphs and solves QA through tool-using agent planning.
- ▶ **Evidence:** Improvements appear in accuracy, citation precision/recall, human groundedness, and noisy settings.

Reasoning Loop



Why Planning Helps



- ▶ Factual queries can terminate quickly with semantic search.
- ▶ Inferential and temporal questions trigger deeper graph traversal.
- ▶ Acoustic-aware questions selectively access raw audio when transcripts are insufficient.

Self-Correction Case

Query: "What is the team's general attitude to the project deadline and workload?"	
Initial answer	Generic summary: the deadline is tight and members feel urgency.
Reflection	Low confidence; needs concrete evidence for "tight" and "work-intensive".
Re-plan	Search for deadline and workload sentiment; retrieve specific utterances.
Final answer	Cites "six days", "change the deadlines", "a hassle", and "superficially".

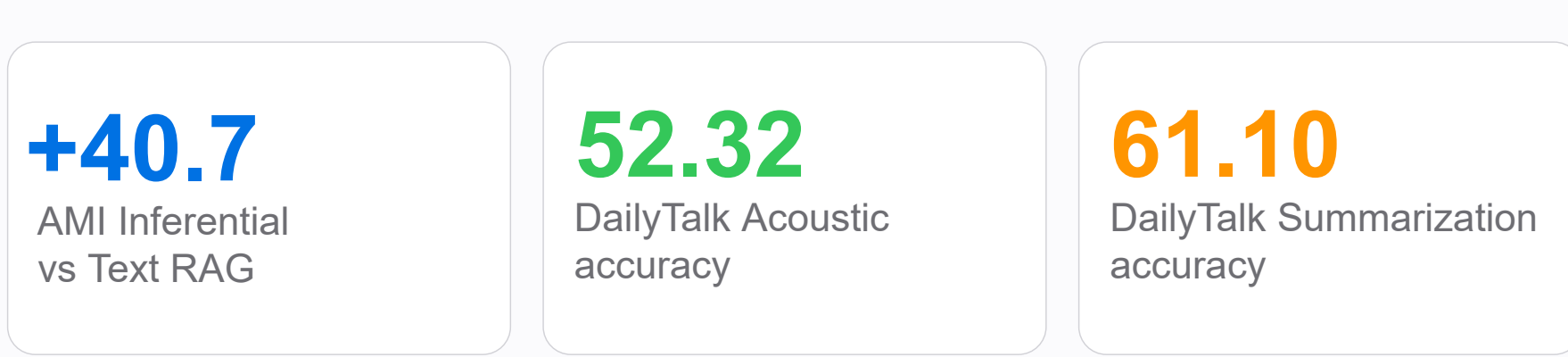
- ▶ Reflection converts unsupported fluent text into evidence-seeking behavior.
- ▶ Final response is anchored to exact utterance-level citations.

Ablation Highlights

Removed Component	Avg.	Drop
Full GRGA	49.49	—
w/o Reflection	39.64	-9.84
w/o Graph Traversal	38.21	-11.27
w/o Audio Access	34.71	-14.78
w/o Semantic Search	16.28	-33.21

- ▶ Semantic search finds entry evidence; graph traversal connects dispersed clues.
- ▶ Audio access is crucial for acoustic-aware questions.
- ▶ Reflection prevents accepting unsupported first-pass answers.

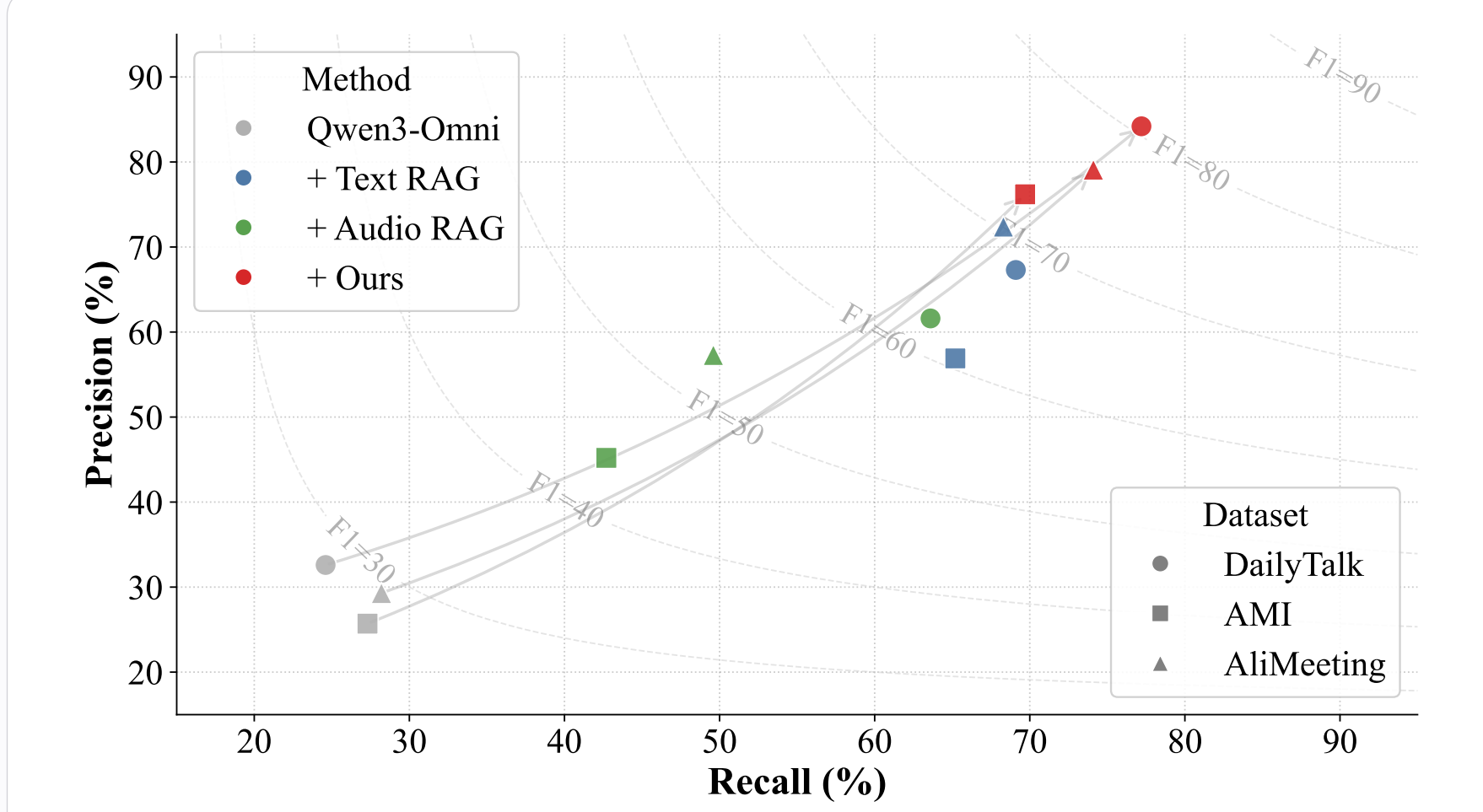
Main Results



AMI Method	Fact.	Infer.	Temp.	Summ.	Accu.
Qwen3-Omni speech	20.96	26.38	15.12	16.67	12.15
MiMo-Audio text	53.62	44.10	37.14	41.49	17.94
BGE-M3 Text RAG	32.56	24.56	23.78	25.36	25.34
CLASP Audio RAG	24.12	20.31	22.76	11.60	14.79
GRGA	59.46	65.29	38.56	48.46	35.68

Semantic Accuracy (%) with LLM-as-a-judge. GRGA is best across all AMI question types.

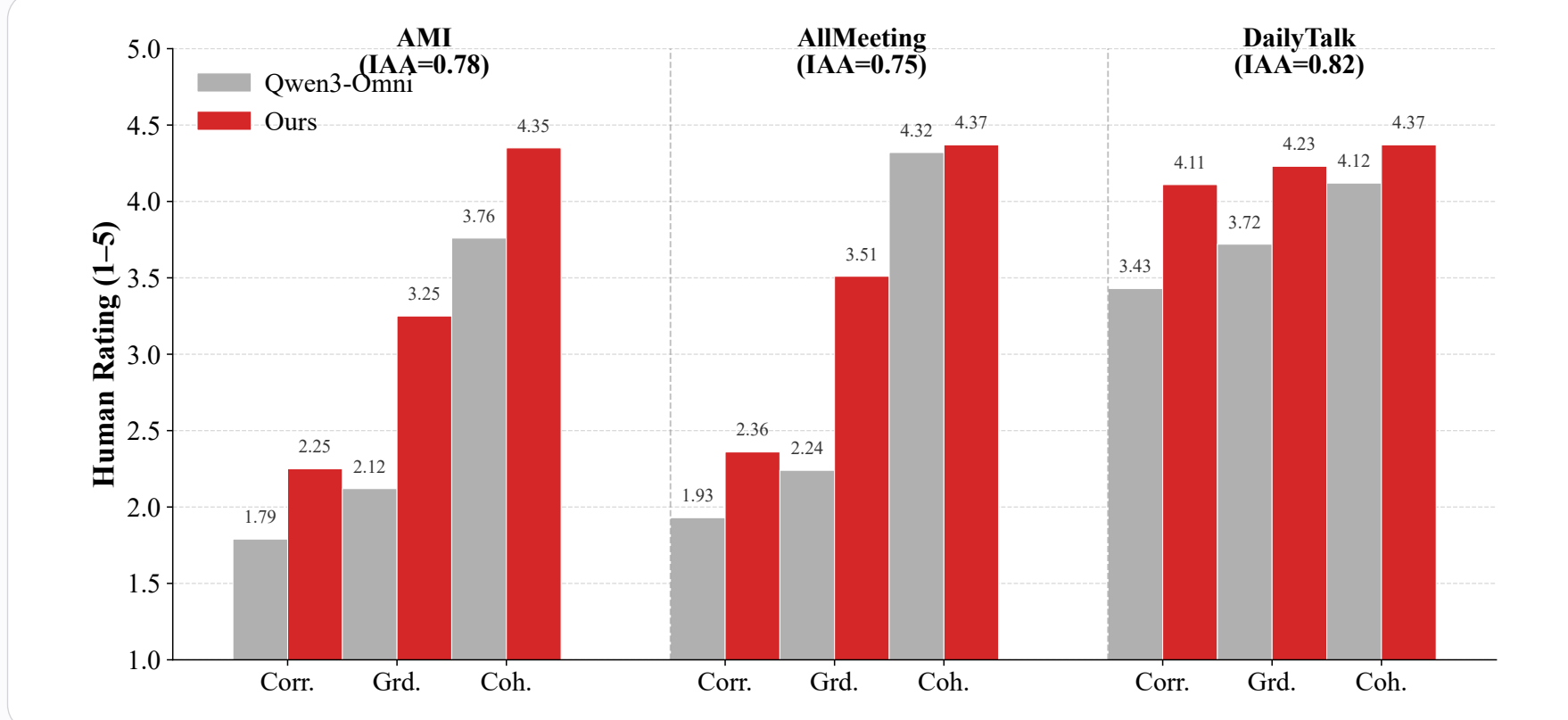
Grounded Answers



Method on AMI	Precision	Recall
Qwen3-Omni	25.7	27.3
+ Text RAG	56.9	65.2
+ Audio RAG	45.2	42.7
+ GRGA	76.2	69.7

Citation accuracy uses a $\pm 2s$ tolerance over timestamp evidence.

Human Evaluation



+0.97
Overall groundedness gain

+0.52
Overall correctness gain